Week 5 Feasibility Memo

AI-Assisted Triage Using the Yale Emergency Department Dataset

The Yale Emergency Department triage dataset appears suitable for developing a baseline AI model to predict Emergency Severity Index (ESI), provided that data cleaning is completed and outcome leakage variables are excluded.

Dataset Summary (Plain Numbers)

* Total patient records: **________**
* Total variables (columns): **________**
* Target variable: **ESI (1–5)**
* Number of vital sign variables: **7**
* Number of chief complaint variables: **Approximately 200**
* Demographic variables include age, gender, race, ethnicity, language, insurance status, and marital status.

The dataset contains structured clinical information collected during emergency department triage and is appropriate for supervised machine learning after preprocessing.

Top Three Reasons the Dataset is Feasible

Reason 1 – Large Structured Dataset

The dataset contains a large number of patient encounters and structured variables collected during triage. These provide sufficient information for training and evaluating predictive models.

Evidence:

* Large number of observations
* Standardised triage measurements
* Multiple clinical variables recorded for each patient

Reason 2 – Relevant Clinical Features

Important predictors such as age, heart rate, blood pressure, respiratory rate, oxygen saturation, temperature, glucose, demographics, and chief complaints are available before clinical decisions are made.

Evidence:

* Exploratory analysis showed meaningful variation across ESI classes.
* Vital signs demonstrate relationships with patient acuity.

Reason 3 – Data Can Be Successfully Cleaned

Most missing values can be handled using standard preprocessing techniques including median imputation for numeric variables and "Unknown" categories for missing demographic fields.

Evidence:

* Missingness analysis identified manageable levels of incomplete data.
* Implausible physiological values can be removed using clinical reference ranges.

Top Three Concerns and Their Mitigation

Concern 1 – Missing Data

Some variables contain missing values that may reduce model performance.

Mitigation

- Median imputation for numerical variables.
- Fill missing categorical values with "Unknown."

Concern 2 – Data Leakage

Variables such as discharge disposition are only known after treatment and would artificially inflate prediction accuracy.

Mitigation

- Remove all outcome variables before model training.
- Restrict predictors to information available during triage.

Concern 3 – Class Imbalance and Bias

Some ESI categories and demographic groups are more common than others, which could bias predictions.

Mitigation

- Monitor class distributions.
- Evaluate performance across demographic groups.
- Consider class weighting if required.

Caveats

Although this dataset is suitable for developing a baseline model, it originates from a single emergency department in the United States.

Clinical workflows, patient demographics, disease prevalence, and available resources differ from hospitals in the Caribbean. Consequently, a model trained on this dataset may not generalise well to Caribbean emergency departments without external validation and local retraining.

Top-10 Feature Shortlist

1. Age
2. Heart rate
3. Systolic blood pressure
4. Respiratory rate
5. Oxygen saturation
6. Temperature
7. Blood glucose
8. Arrival mode
9. Chief complaint indicators
10. Gender

Outcome variables such as disposition and previous disposition are intentionally excluded because they represent information unavailable at the time of triage.

Conclusion

Overall, the exploratory analysis indicates that the dataset is appropriate for developing an initial AI-assisted triage model. After cleaning the data, handling missing values, removing leakage variables, and selecting clinically relevant predictors, the dataset is suitable for training baseline machine learning models such as logistic regression and decision trees.

The next step is to build and evaluate these baseline models using the cleaned dataset and the selected predictor variables.